# KIQ 1.4 — Patent IPC Co-occurrence Network Insight Recap

**Key Intelligence Question (KIQ 1.4):** What are the main strengths and weaknesses of METRO’s digital ordering channels from a customer perspective?

**Core Indicators:** Positive/negative app-store review themes; complaints about usability or bugs; praise for convenience, speed, or product range.

### Patent Corpus Used

A targeted patent corpus of approximately 250 recent patents (2022–2026) related to user interfaces, automated application testing, and digital store customer feedback loops was extracted from Espacenet.

### Method Applied

IPC co-occurrence mapping (inter-class links) structured as a semantic network was used to identify how technological strengths and solutions to technical bugs are interconnected.

### Main Insights (Patent IPC Co-occurrence Network)

- Transaction-to-Interface Links: Strong co-occurrence links between G06Q30/06 (ordering) and G06F3/048 (interface) reveal that commercial convenience cannot be separated from interface simplicity. A breakdown in interface layout directly causes order cancellation.
- Bug Mitigation Clusters: Substantial cross-class connections exist between data processing (G06F16/00) and error logging (G06F11/00), indicating that modern architectures link catalog speed with background crash resolution systems.
- Feedback Integration: Network nodes show an emerging pattern of connecting real-time user session monitoring with predictive layout changes to automatically address customer complaints.

### Strategic Interpretation

The structural linkage map demonstrates that platform convenience (speed) is technically dependent on background data management. METRO must ensure its architecture tightly links product range rendering speed with efficient error handling systems to minimize negative reviews.